Supplementary Table 1 - Metadata of Related Papers

Citation	Scientific Journal	Title	Year	Methodology	Architecture	Debris Depth	Dataset Used	Class Count	Class Info	Data Count	Results	Hardware
Kylili, K., et al. (2019)	Environmental Science and Pollution Research (Q1)	Identifying floating plastic marine debris using a deep learning approach	2019	Detection	VGG-16	Surface Debris	ImageNet. Collected.	3	Bottles, buckets, straws	12,000 images	Validation accuracy 86%	Intel Xeon (CPU 2.40 GHz) processor with NVIDIA (Quadro K4200) graphics card
Fulton, M., et al. (2019)	2019 International Conference on Robotics and Automation (ICRA)	Robotic Detection of Marine Litter Using Deep Visual Detection Models	2019	Object Detection	TinyYOLO, YOLOv2, SSD, Faster RCNN	Underwater	Trash-ICRA19	3	Plastic, ROV, Bio	5,720 images	Faster RCNN mAP: 81%	GPU (NVIDIA GTX 1080); Embedded GPU (NVIDIA Jetson TX2); CPU (Intel i3-6100U)
Watanabe, J., et al. (2019)	Journal of Applied Remote Sensing (Q2)	Underwater and airborne monitoring of marine ecosystems and debris	2019	Object Detection/Airborne monitoring	YOLOv3	Underwater	Collected & Google Open Images	3;4	Fish, turtles, jellies; plastic bags, plastic bottles, other debris, driftwood	189 (debris) 8,036 (biodiversity)	mAP Sea life 69.6% Debris 77.2%	Intel® CoreTM i7-7800X CPU 3.50 GHz, 64 bits, 40-GB RAM, GPU Nvidia GTX 1080, CUDA 9.0, cuDNN 7.0.3, and OS ubuntu 16.04.
Kylili, K., et al. (2020)	Environmental Science and Pollution Research (Q1)	An intelligent way for discerning plastics at the shorelines and the seas	2020	Classification	VGG16	Surface Debris	ImageNet. Algalita. Collected through use of cameras	8	Plastic bottles, plastic buckets, plastic bags, fishing nets, plastic straws, and food wrappings. Flying fish & Other	1,600 raw; 32,000 images	90% val accuracy	Intel® Xeon® machine equipped with an Intel CPU Core E5-2630 v3 (2.40 GHz) with 48.0 GB of memory (RAM) and an NVIDIA Quadro K4200 graphics card, clock-rated at 784 MHz with 28.6 GB of memory.
Panwar, H., et al. (2020)	Case Studies in Chemical and Environmental Engineering (Q1)	AquaVision: Automating the detection of waste in water bodies using deep transfer learning	2020	Dataset/Detection	RetinaNet (Resnet50 as backbone & FPN)	Underwater	TACO dataset (JEDI)	4	Plastic; Glass; Paper; Metal	369 images	mAP 81%	-
Kylili, K., et al. (2021)	Marine Pollution Bulletin (Q1)	A new paradigm for estimating the prevalence of plastic litter in the marine environment	2021	Object Detection & Instance Segmentation	YOLOV5x & YOLACT++	Shore Debris	ImageNet. Algalita.	7	Bottles; Bags; Buckets; Fishing nets; Straws; Fish; Wrappers	1,650 images	AP 92.4%; 69.6%	NVIDIA® Tesla® K80 graphical processing unit (GPU)
Xue, B., et al. (2021b)	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (Q1)	An Efficient Deep-Sea Debris Detection Method Using Deep Neural Networks	2021	Object Detection	ResNet50-YOLOv3	Underwater	JEDI	7	Fishing net & rope, metal, cloth, glass, rubber, plastic, and natural debris	10,000 images	mAP 83.4%	GeForce GTX 1080Ti GPU with- a capacity of 11 GB
Deng, H., et al. (2021)	Sensors (Q1)	An Embeddable Algorithm for Automatic Garbage Detection Based on Complex Marine Environment	2021	Instance Segmentation/ Object Detection	Mask RCNN	Underwater	TrashCan 1.0	22	TrashCan-Instance	7,212 images	mAP 59.2%; 65.2%	Ubuntu18.04. Intel(R) Xeon(R)Silver 4110 CPU @2.10GHz. GeForce RTX 2080Ti. CUDA 10.2.
Xue, B., et al. (2021a)	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (Q1)	Deep-Sea Debris Identification Using Deep Convolutional Neural Networks	2021	Object Detection	Shuffle-Xception	Underwater	JEDI	7	Fishing net & rope, metal, cloth, glass, rubber, plastic, and natural debris	13,914 images	95% precision average	GeForce GTX 1080Ti GPU and installed with Intel(R) Xeon(R) W-2133 CPU at 3.60 GHz, 31.7 GB RAM,
Hipolito, J. C., et al. (2021)	IEEE 19th Student Conference on Research and Development (SCOReD)	Detection of Underwater Marine Plastic Debris Using an Augmented Low Sample Size Dataset for Machine Vision System: A Deep Transfer Learning Approach	2021	Object Detection	YOLOv3	Underwater	Dataset of Underwater Trash (sample)	1	Non Bio	300 images	mAP 98.15%	-
Teng, C., et al. (2022)	Marine Pollution Bulletin (Q1)	Deploying deep learning to estimate the abundance of marine debris from video footage	2022	Object Detection & Tracking	YOLOv5	Surface Debris	ImageNet. Algalita. Collected. Pretrained MSCOCO	9	Plastic bottles, Plastic buckets, Plastic bags, Fishing nets, Plastic straws, Food wrappings, A fish species, Aluminum cans, Cigarette butts	2,050 images	mAP 89.4 %	NVIDIA® Tesla® K4 GPU from Google Colab
Moorton, Z., et al. (2022)	Marine Pollution Bulletin (Q1)	Is the use of deep learning an appropriate means to locate debris in the ocean without harming aquatic wildlife?	2022	Classification	VGG-16 & custom	Underwater	Collected (JEDI incl)	2	Animal, Litter	1,744 images	accuracy of 89% CNN and 95% VGG-16	Dell Inspiron i7-7700HQ CPU 2.8GHz, 16GB RAM, 64-Bit with a Nvidia GeForce GTX 1050 Ti graphics card, running on Windows 10.
Kikaki, K., et al. (2022)	PloS one (Q1)	MARIDA: A benchmark for Marine Debris detection from Sentinel-2 remote sensing data	2022	Dataset/Remote Sensing	Random Forest model (RF) & U-Net architectuure	Surface Debris	Marine Debris Archive (MARIDA)	15	Marine Debris, Dense Sargassum, Sparse Sargassum, natural organic material, ship, clouds, marine water, sediment-laden water, foam, turbid water, shallow water, waves, cloud shadows, wakes, mixed water	1,381 patches; 3,399 marine debris pixels	92%; 70% accuracy	-
Sánchez-Ferrer, A., et al. (2022)	Iberian Conference on Pattern Recognition and Image Analysis	The CleanSea Set: A Benchmark Corpus for Underwater Debris Detection and Recognition	2022	Dataset/ Instance Segmentation	Mask R-CNN	Underwater	CleanSea Set	19	Plastic bag; Bumper; Metal waste; Can; Wood; Tire; Packaging; Towel; Shoe; Bottle; Glove; Metal chain; Fishing net; Pipe; Washer; Plastic waste; Basket; Rope; Square can	1,223 images	mAP 60%	Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz with 16 GB RAM, a NVIDIA GeForce RTX 2070 with 6 GB GDDR6 Graphics Processing Unit (GPU) with the cuDNN library.
Sánchez-Ferrer, A., et al. (2023)	Pattern Recognition Letters (Q1)	An Experimental Study on Marine Debris Location and Recognition Using Object Detection	2023	Instance Segmentation	Mask R-CNN	Underwater	CleanSea Set	17	Plastic bag; Bumper; Metal waste; Can; Wood; Tire; Packaging; Towel; Shoe; Bottle; Glove; Fishing Net; Pipe; Plastic Waste; Basket; Rope; Square can	1,223 images	mAP Instance 63.5%; Material 65.2%	-
Liu, J., & Zhou, Y. (2023)	Neural Networks, Information and Communication Engineering (NNICE), International Conference	Marine debris detection model based on the improved YOLOv5	2023	Object Detection	YOLOv5s modified (MobileNetv3 backbone)	Underwater	TrashCan 1.0	50	TrashCanMaterial; TrashCan-Instance	7,212 images	mAP 67%	*Intel(R) Xeon(R) Silver 4210R CPU@2.20GHz with an NVIDIA GeForce RTX 2090Ti GPU.
Zhou, W., et al. (2023)	IEEE Transactions on Instrumentation and Measurement (Q1)	YOLOTrashCan: A Deep Learning Marine Debris Detection Network	2023	Object Detection	YOLOTrashCan	Underwater	TrashCan 1.0	16; 22	TrashCanMaterial; TrashCan-Instance	7,212 images	mAP 58.66%; 65.01%	AMD Ryzen 7 3700X, Nvidia TITAN RTX 24 GB, and RAM 48G
Jain, R., et al. (2024)	2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS)	Advancing Underwater Trash Detection: Harnessing Mask R-CNN, YOLOv8, EfficientDet-D0 and YOLACT	2024	Object Detection	YOLOv8	Underwater	TrashCan 1.0	22	TrashCan-Instance	7,212 images	mAP 71%	Tesla P100 GPU as an accelerator.
Jiang, W., et al. (2024)	Journal of Marine Science and Engineering (Q2)	Research on the Identification and Classification of Marine Debris Based on Improved YOLOv8	2024	Object Detection	YOLOv8 modified	Underwater	TrashCan 1.0	16; 22	TrashCanMaterial; TrashCan-Instance	7,212 images	mAP 66.7%; 72%	Python 3.7, PyTorch 1.10, CUDA 10.2, and CUDNN 7.6, using a 24-GB NVIDIA GeForce RTX 3090 equipped with Windows 10
Đuraš, A., et al. (2024)	Scientific Data (Q1)	A Dataset for Detection and Segmentation of Underwater Marine Debris in Shallow Waters	2024	Object Detection	Faster RCNN & YOLOv6	Shallow Water	Seaclear Marine Debris Dataset	40	3 Super Categories: Debris, bio, robot	8,610 images	mAP 61.7%; 68.3%	BlueROV2: Bluerobotics Low-Light HD Camera and a Paralenz Vaquita; SST Mini-Tortuga ROV: Smart Security SIP-E323CV.
Shen, A., et al. (2024)	IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	Marine Debris Detection in Satellite Surveillance Using Attention Mechanisms	2024	Remote Sensing/Instance Segmentation/Object Detection	CBAM (enhanced YOLOv7 with attention backbone)	Surface/Shallow	Collected from PlanetScope	2	with debris', 'without debris'	321 images	mAP@50: 76%; 72%	Google Colab
Huang, C., et al. (2025)	IEEE Access (Q1)	YOLO-MES: An Effective Lightweight Underwater Garbage Detection Scheme for Marine Ecosystems	2025	Object Detection	YOLO-MES	Underwater	J-EDI (JAMSTEC)	5	Debris, metal, plastic, rubber, net-rope	6,283 images	95.8% accuracy	Intel(R) Core(TM) i9 − 11900 CPU @ 2.50 GHz, 64GB RAM, Nvidia GeForce RTX A4000 GPU. CUDA 11.3
Zhao, F., et al. (2025)	Marine Pollution Bulletin (Q1)	Seafloor debris detection using underwater images and deep learning-driven image restoration: A case study from Koh Tao, Thailand	2025	Object Detection	SFD-YOLO (enhanced YOLOv8)	Seafloor	Collected. Pretrained on TrashCan 1.0	11	Plastic bags, bottles, sheets, solid objects, boxes, metal sheets, tubes, glass bottles, fishing nets, ropes and cans	294 patch images from video	mAP 91.2%	RTX 4090 GPU with 64 GB of RAM. 4k GoPro.
Ma, J., et al. (2025)	Marine Pollution Bulletin (Q1)	Toward smart ocean monitoring: Real-time detection of marine litter using YOLOv12 in support of pollution mitigation	2025	Object Detection	YOLOv12	Floating and Underwater	Garbage in the Ocean dataset	15	Mask, glove, metal, net, gbottle, cellphone, pbag, pbottle, plastic, rod, sunglasses, tire, can, electronics, and misc	5,130 images	mAP@50:84%; mAP@50–95: 70%	-
Walia, J. S., et al. (2025).	IEEE Access (Q1)	Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis	2025	Object Detection	YOLOv5; YOLOv7; YOLOv8	Underwater	Underwater Trash Detection dataset	3	Underwater Trash, Rover, Biological life.	10,000 images	mAP 96%; 96%; 96%	Tesla T4 GPU
												*RTX 2090 Ti does not exist - we can only assume this is a typo instead of 2080ti or 3090